FRONT-END DEVELOPMENT

ASSIGNMENT 3 TEMPLATE

STUDENT NAME: RANJITH M

YEAR: II YEAR

COLLEGE CODE: 201

COLLEGE NAME: ARIGAR ANNA Government ARTS & SCIENCE COLLEGE

DEPARTMENT: B.Sc Computer Science

SEMESTER: IV

ROLL NO: 24CSC124

NMID: asanm20124132011802521080

DATE: 10/03/2026

PROJECT TITLE: Front-End Development (Simple Event Registration Page)

1. PROBLEM STATEMENT:

The task is to enhance the event registration page (built earlier with HTML and styled with CSS) by adding JavaScript interactivity. This involves validating user inputs, preventing invalid submissions, and dynamically displaying registered participants — all on the client side, without using any backend or database.

2. OBJECTIVE:

- Add form validation to ensure only correct data is submitted.
- Provide real-time feedback to users (valid/invalid inputs).
- Dynamically update the participants table with submitted data.
- Improve user experience by preventing page reloads and showing success messages.
- Reset the form after submission while keeping the table intact.

3 REQUIREMENTS:

3.1 Form Validation:

- Name → Cannot be empty.
- Email → Must follow proper format (example@domain.com).
- Phone Number → Exactly 10 digits.
- Date of Birth → Must be selected.
- Gender → Must be chosen.
- Event Selection → Must select an event.
- Mode of Participation → Must be selected.

3.2 Confirmation Checkbox Validation:

Use JavaScript if conditions and regex for email/phone. Apply CSS classes (valid/invalid) for visual feedback.

Ensure the declaration checkbox is ticked before submission.

If unchecked, block submission and show an error message.

3.3.Dynamic Display of Registered Participants:

Capture form data using DOM methods (get Element By Id, query Selector).

- Create a new table row () and cells () with participant details.
- Append the row to the participants table dynamically.

3.4 Prevent Page Reload:

- *Use event .prevent Default() in the form's on submit handler.*

- *This ensures the page doesn't refresh and data updates instantly.*

3.5 Clear Form After Successful Submission:

- *Call form. reset() after adding participant details.*

- *Keeps table data intact while clearing input fields*

3.6 Display Success Message:

- *Show a message like **"Registration Successful!"** using inner HTML.*

- *Use set Timeout() to remove the message after a few seconds.*

3.7 Input Field Feedback:

- *Highlight valid fields in **green** and invalid ones in **red**.*

Example CSS:
.valid { border: 2px solid green; } .invalid { border: 2px solid red; }

.valid { border: 2px solid green; }
.invalid { border: 2px solid red; }

3.8 Optional Enhancements:

- Participant Count → *Show total registrations.*
- *Delete Row Option → Allow removing participants.*
- *Duplicate Email Prevention → Check if email already exists before adding.*
- *Event Summary → Display participants grouped by event.*

4 JAVASCRIPT CONCEPTS USED:

- DOM Selection & Manipulation (get Element By Id, create Element, append Child).
- Event Handling (on submit, onclick).
- Validation Logic (regex, conditionals).
- Dynamic Element Creation (adding rows to table).
- Timers (set Timeout for temporary messages).

5 IMPLEMENTATION STEPS:

1. *Create the HTML form with all required fields (name, email, phone, DOB, gender, event, mode, checkbox).*
2. *Add a participants table below the form to show registered users.*
3. *Write JavaScript and connect it to the form.*
4. *Use prevent Default() to stop page reload when submitting.*
5. *Validate inputs: check name not empty, email format, phone = 10 digits, DOB/gender/event/mode selected, checkbox ticked.*

6 OUTPUT:

Web Development Workshop 2026

This workshop helps students learn HTML, CSS and JavaScript basics.

Event Information

Date: 15 March 2026
Time: 10:00 AM – 4:00 PM
Venue: Seminar Hall A
Organizer: Department of Computer Science

Personal Information

Name:

Email:

Phone:

Date of Birth:

dd - mm - yyyy

Gender:
○ Male ○ Female

Event Selection

Select Event:

--Select--

Mode:
○ Online ○ Offline

Additional Details

College/Company:

Upload ID Proof:
Choose File No file chosen

Comments:

☐ I confirm that the above details are correct.

Submit Reset

Registered Participants

Name	Email	Phone	Event	Mode
Rahul Kumar	rahul@gmail.com	9876543210	Web Development Workshop	Offline
Priya Sharma	priya@gmail.com	9123456780	AI Seminar	Online
Arjun Patel	arjun@gmail.com	9988776655	Cyber Security Bootcamp	Offline

Contact Email: events@abccollege.edu
Help Desk: +91 94444 12345
© 2026 Department of Computer Science

7 LEARNING OUTCOME:

By completing this assignment, you will:

- *Master form validation techniques.*
- *Gain experience in DOM manipulation.*
- *Learn to build interactive web applications using JavaScript.*
- *Understand how to enhance user experience with dynamic updates and feedback mechanisms.*

8 CONCLUSIONS:

By implementing JavaScript interactivity, the event registration page becomes dynamic, user-friendly, and responsive**.** Validation ensures that only correct data is submitted, while features like success messages, form reset, and real-time feedback improve the overall experience. The participants table updates instantly without page reload, making the application feel smooth and professional.